Unit 12

Vector-Valued Functions

In 2D space (plane)

$r(t) = f(t)i + g(t)j$

or $r(t) = \langle f(t), g(t) \rangle$

In 3D

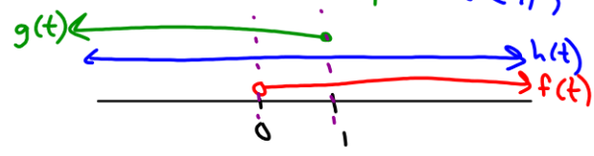
$r(t) = f(t)i + g(t)j + h(t)k$

or $r(t) = \langle f(t), g(t), h(t) \rangle$

A vector-valued function feeds t into component functions: r(t) = f(t)i + g(t)j in the plane, and adds h(t)k in space. Each r(t) is a position vector.

$r(t) = \ln(t)\,i + \sqrt{1-t}\,j + t\,k$

$t > 0 \quad t \le 1 \quad t \in \mathbb{R}$



$D: \; 0 < t \le 1$

The domain is where every component is defined: ln t needs t > 0, √(1−t) needs t ≤ 1, and t works for all t. Overlap them: 0 < t ≤ 1.

$r(t) = 2\cos(t)\,i - 3\sin(t)\,j$

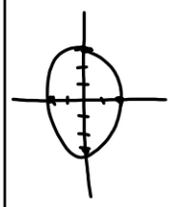
$0 < t < 2\pi$

$2\cos(t) = x$

$-3\sin(t) = y$

$\sin^2(t) + \cos^2(t) = 1$

$\left(\frac{y}{-3}\right)^2 + \left(\frac{x}{2}\right)^2 = 1$



$\frac{y^2}{9} + \frac{x^2}{4} = 1$

To sketch, eliminate t. With x = 2cos t and y = −3sin t, use sin²t + cos²t = 1 to get x²/4 + y²/9 = 1, an ellipse.

$r(t) = 4\cos(t)\,i + 4\sin(t)\,j + t\,k$

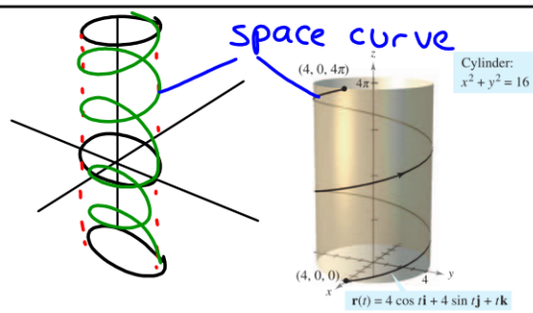
$x = 4\cos(t) \qquad y = 4\sin(t) \qquad z = t$

$\left(\frac{x}{4}\right)^2 + \left(\frac{y}{4}\right)^2 = 1$

$\frac{x^2}{16} + \frac{y^2}{16} = 1$

$x^2 + y^2 = 16$

Here x = 4cos t and y = 4sin t give x² + y² = 16, while z = t keeps climbing. The curve wraps around that cylinder: a helix.



The helix r(t) = 4cos t i + 4sin t j + t k is a space curve: it winds up the cylinder x² + y² = 16, rising as t increases.

$y = x^2 \longrightarrow r(t) = t\,i + t^2\,j$

$x = t \rightarrow y = t^2$

Any graph y = f(x) becomes a vector function by letting x = t: y = x² turns into r(t) = t i + t² j.

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$\frac{x^2}{12}+\frac{y^2}{24}+\frac{z^2}{4}=1$

$y=x^2$ let $x=t$ $y=t^2$

$z\geq 0$

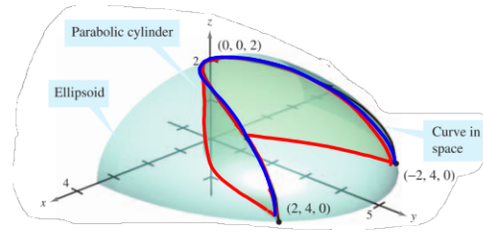
→ solve for z $2x^2+y^2+6z^2=24$

$z=\pm\sqrt{\frac{24-2x^2-y^2}{6}}$

$z=\sqrt{\frac{24-2t^2-t^4}{6}}$

$C(t)=t\,i+t^2\,j+\sqrt{\frac{24-t^2-t^4}{6}}\,k$

Curve of intersection: let x = t, so y = t². Clear fractions (2x² + y² + 6z² = 24) and solve for z ≥ 0: z = √((24 − 2t² − t⁴)/6). Then C(t) = t i + t² j + z k.



The picture: the parabolic cylinder y = x² slices through the ellipsoid, and the edge where they meet is the space curve C(t) from the previous slide.

$\lim_{t\to a} r(t)=\lim_{t\to a} f(t)\,i+\lim_{t\to a} g(t)\,j+\lim_{t\to a} h(t)\,k$

$r^{(n)}(t)=f^{(n)}(t)\,i+g^{(n)}(t)\,j+h^{(n)}(t)\,k$

$\int r(t)\,dt=\int f(t)\,dt\,i+\int g(t)\,dt\,j+\int h(t)\,dt\,k$

Limits, derivatives and integrals all work component by component: apply the operation to f, g and h separately and keep the i, j, k.

$\int_0^1 r(t)\,dt \qquad r(t)=\sqrt[3]{t}\,i+\frac{1}{t+1}\,j+e^{-t}\,k$

$R(t)=\frac{3}{4}\,i+\ln(2)\,j+\left(1-\frac{1}{e}\right)k$

Integrate each component from 0 to 1: ∫t^(1/3) dt = 3/4, ∫1/(t+1) dt = ln 2, ∫e^(−t) dt = 1 − 1/e. Answer: (3/4)i + ln 2 j + (1 − 1/e)k.

$r'(t)=\cos(2t)\,i-2\sin(t)\,j+\frac{1}{1+t^2}\,k$

$r(0)=-3\,i+2\,j+k$

Find $r(t)=\left\langle \frac{1}{2}\sin(2t)+C_1,\ 2\cos(t)+C_2,\ \arctan(t)+C_3\right\rangle$

$C_1=3 \qquad C_2=0 \qquad C_3=1$

$r(t)=\left(\frac{1}{2}\sin(2t)+3\right)i+2\cos(t)\,j+\left(\arctan(t)+1\right)k$

Antidifferentiate each piece, then use r(0) = 3i + 2j + k to find the constants C₁ = 3, C₂ = 0, C₃ = 1: r(t) = (½sin 2t + 3)i + 2cos t j + (arctan t + 1)k.

$r(t)=\cos(t)\,i+\sin(t)\,j+2t\,k$

$r'(t)\cdot r''(t)=0$

For r(t) = cos t i + sin t j + 2t k: r′ = ⟨−sin t, cos t, 2⟩ and r″ = ⟨−cos t, −sin t, 0⟩, so r′·r″ = 0. The speed ‖r′‖ = √5 never changes.

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